HANANEEL EPHRAIM MANDAM

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EDUCATION

B.E. Mechanical Engineering | BITS Pilani, Hyderabad Campus *Expected 2028*

Coursework: Artificial Intelligence, Vibrations & Control, Mechanisms & Machines, Advanced Mechanics of Solids, Fluid Mechanics, Applied Thermodynamics, Principles of Aerodynamics, Heat Transfer, Differential Equations

EXPERIENCE

Research Intern, Physiological Signal Processing | CSIR-CEERI, Chennai *May 2026 – Jul 2026*

- Built an end-to-end real-time ECG/HRV pipeline that ingests commercial health-band data and estimates KSS drowsiness levels over live rolling windows, for a defense project.
- Improved Pan-Tompkins++ R-peak detection to **98.8%** accuracy on the MIT-BIH NSTDB benchmark (24 dB and 18 dB SNR), vs. **81.7%** for NeuroKit2.
- Designed a pub/sub ECGBus architecture for real-time multi-subject BLE session management, with hardware-level signal quality gating on Polar H10 straps and automated logging of vehicle and software crash events.

Powertrain Engineer & Electrical Lead | Electric Baja SAE, BITS Pilani Hyderabad *Apr 2025 – Apr 2026*

- Designed a 4:1 inline spur gear reduction and chain-sprocket drivetrain in SolidWorks; built free-body diagrams of gear and shaft loads, sized gears with Lewis and AGMA bending and contact-stress hand calcs, then validated shaft and gear geometry under **90 Nm** peak torque with finite element analysis (FEA) in ANSYS.
- Built the vehicle safety interlock from scratch to SAE E-Baja rules: microcontroller code, relay logic, and the HV/LV interface; validated shutdown against multiple failure cases and designed the full wiring schematic.
- Retrofitted chassis members via profiling and welding for the electric drivetrain conversion; coordinated a **7-person** team through fabrication and vendor timelines.

PROJECTS

Uncertainty-Aware UAV Emergency Landing Zone Selection | GitHub *Jun 2026 – Present*

- Built a three-channel risk fusion pipeline: semantic segmentation (SegFormer-B2 fine-tuned on UAVid), epistemic uncertainty via MC Dropout, and temporal label-disagreement tracking.
- Ablations on UAVid and Aeroscapes: temporal consistency alone still marked unsafe zones as safe on both datasets; every uncertainty-aware variant produced **zero** false-safe selections.

Ground-Effect Wing CFD | Principles of Aerodynamics Coursework *In Progress*

- Built a 2D RANS workflow (k-omega SST) in ANSYS Fluent for an inverted NACA 6412 wing; swept angle of attack from -12° to +4° to map downforce, drag, and stall.
- Linear-range lift slope of **0.101/deg** and zero-lift angle within **0.3°** of thin-airfoil theory; peak L/D of **62**; ride-height study next.

Self-Landing EDF Hopper: TVC Gimbal & Roll Vanes | Personal Project *In Progress*

- Designed a two-axis thrust vector control gimbal in SolidWorks for a 70 mm EDF: ±10° on both axes via a ~3:1 servo pushrod linkage, plus two direct-drive NACA 0012 roll vanes sized to the jet core.
- Ran an 11-configuration interference study that found two tab collisions; traced them to tab corners swinging out at tilt and fixed them (~1.5 mm clearance at 10°). Resolved a pitch-yaw linkage bind with an hourglass hole before printing.
- Moved servos above the gimbal to clear the exhaust; bench-tested actuation on an ESP32. Next: IMU-driven gimbal demo, then load-cell thrust and roll-authority tests.

Image Stitching for Vision-Based Dimensional Inspection | ME F376 Design Project *Aug 2026 – Present*

- Built an image-stitching pipeline in Python/OpenCV from first principles (ORB, RANSAC/MAGSAC++ homography and similarity fits) to inspect parts too large for one frame; picks alignments by pixel agreement (NCC), not inlier count.
- Validated on an 84-case ground-truth suite: **83/84** pass (< 5 px error), refusing rather than guessing outside its envelope; the suite caught a **320 px** silent misalignment on repeating patterns and a regression from its first fix.

SKILLS

Programming & Tools: Python (PyTorch, NumPy, OpenCV, HuggingFace), MATLAB, Simulink, C, Java, Git, Linux, Excel

CAD, FEA & CFD: SolidWorks, ANSYS (Static Structural, Modal, Fluent), Finite Element Analysis (FEA), Mechanical Design

Structural Analysis: Free-Body Diagrams, Structural Load Analysis, Shear & Moment Diagrams, Lewis/AGMA Gear Rating

ML & Vision: Semantic Segmentation, Uncertainty Quantification, Transfer Learning, Image Registration

Fabrication: MIG/TIG Welding, Lathe, Sheet Metal, FDM 3D Printing, Soldering